T3000 API & Protocol

Message: UPDATE_WEBVIEW_LIST · built Nov 13, 2026

Message: UPDATE_WEBVIEW_LIST

UPDATE_WEBVIEW_LIST writes values to device entries in bulk. Use this to update multiple inputs, outputs, variables, or other entry types in a single operation.

Common Uses:

- Updating output values
- Setting multiple setpoints
- Batch configuration changes
- Program control (start/stop)

Overview

Message Type: UPDATE_WEBVIEW_LIST **File Location:** `BacnetWebView.cpp` line 1957 **Purpose:** Write multiple entry values to device

Request Format

JSON

```
{
  "action": 16,
  "source": 1,
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 2,
  "objectinstance": 123456,
  "data": [
    {
      "index": 0,
      "value": 500
    },
    {
      "index": 1,
      "value": 750
    }
  ]
}
```

Parameters

Parameter	Type	Required	Description
`action`	Integer	✅	Must be 16
`source`	Integer	✅	Message source: 0=T3000, 1=WebUI
`panelId`	Integer	✅	Panel number (0-254)
`serialNumber`	Integer	✅	Device serial number
`entryType`	Integer	✅	Type of entry to update
`objectinstance`	Integer	✅	BACnet object instance
`data`	Array	✅	Array of updates (index + values)

Update Data Fields

Varies by entry type. Common fields:

Field	Type	Description
`index`	Integer	Entry index to update
`value`	Number	New value
`auto_manual`	Integer	0=Auto, 1=Manual
`range`	Integer	Unit/range selection

Response Format

JSON

```
{
  "action": "UPDATE_WEBVIEW_LIST_RES",
  "data": {
    "status": true,
    "updated_count": 2,
    "failed_indices": []
  }
}
```

Supported Entry Types

Type	Value	Writable Fields
BAC_OUT	2	value, auto_manual, range
BAC_VAR	3	value, auto_manual
BAC_PRG	6	on_off, auto_manual
BAC_PID	11	setpoint, auto_manual, proportional, reset, rate

Implementation

Write Process

1. **Parse updates** - Extract index and field changes
2. **Validate range** - Check indices are valid
3. **Update global array** - Modify cached data
4. **Write to device** - Send BACnet write command
5. **Verify write** - Confirm success

6. **Update cache file** - Persist changes

Code Pattern

C++

```
case WEBVIEW_MESSAGE_TYPE::UPDATE_WEBVIEW_LIST:
{
    int entry_type = json.get("entryType", Json::nullValue).asInt();
    Json::Value data = json["data"];

    for (int i = 0; i < data.size(); i++)
    {
        int index = data[i].get("index", -1).asInt();

        switch (entry_type)
        {
            case BAC_OUT:
                // Update output value
                if (data[i].isMember("value"))
                {
                    int value = data[i]["value"].asInt();
                    g_Output_data[panel_id].at(index).value = value;
                    WriteOutputValue(objectinstance, index, value);
                }
                break;
            // ... other types ...
        }
    }

    break;
}
```

Error Handling

Invalid Index:

JSON

```
{
  "action": "ERROR_RES",
  "error": {
    "code": -2,
    "message": "Invalid index",
    "details": "Index 70 exceeds max count 64 for outputs"
  }
}
```

Write Timeout:

JSON

```
{
  "action": "ERROR_RES",
  "error": {
    "code": -4,
    "message": "Write failed",
    "details": "Device did not acknowledge write for output 5"
  }
}
```

Performance

Operation	Time
Single value	100-300ms
2-5 values	200-500ms
10+ values	500ms-2s

Usage Examples

Example 1: Update Single Output

JSON

```
{
  "action": 18,
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 2,
  "objectinstance": 123456,
  "data": [
    {
      "index": 0,
      "value": 500,
      "auto_manual": 0
    }
  ]
}
```

Example 2: Update Multiple Variables

JSON

```
{
  "action": 18,
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 3,
  "objectinstance": 123456,
  "data": [
    { "index": 0, "value": 720 },
    { "index": 1, "value": 680 },
    { "index": 2, "value": 750 }
  ]
}
```

Example 3: Start/Stop Programs

JSON

```
{
  "action": 18,
  "panelId": 0,
  "serialNumber": 12345,
  "entryType": 6,
  "objectinstance": 123456,
  "data": [
    { "index": 0, "on_off": 1 },
    { "index": 1, "on_off": 0 }
  ]
}
```

See Also

- UPDATE_ENTRY - Update single field
- GET_WEBVIEW_LIST - Read values
- Control Messages Index